Lawrence Hollom

Institute of Mathematics, EPFL SB MATH, 1015 Lausanne, Switzerland
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Current Position

Postdoctoral Researcher — *February 2026 – Present*
Supervisor: Marius Tiba — *EPFL*

Research Interests

- Extremal and probabilistic combinatorics
- Random graphs and percolation
- High-dimensional geometry
- Infinitary and structural combinatorics

Education

PhD in Pure Mathematics — *September 2022 – January 2026*
Supervisor: Béla Bollobás — *University of Cambridge*

Part III of the Mathematical Tripos — *September 2019 – July 2020*
Master's degree — *University of Cambridge*

Bachelor's degree in Mathematics — *September 2016 – July 2019*
First class, ranked 2nd in year — *University of Cambridge*

Awards

Frontiers of Science Award — *ICBS, 2026*
This prize is awarded at the International Congress of Basic Science (Beijing) to the authors of approximately 75 papers published in the last few years judged to be major advances in the mathematical sciences. The award recognised my work on the bunkbed conjecture.

Smith–Knight and Rayleigh–Knight First Prize — *University of Cambridge, 2024*
Awarded by the University of Cambridge for a submitted essay presenting original mathematical research. The essay received the highest possible prize.

International Mathematical Olympiad Silver Medal — *Hong Kong, 2016*

Previous roles

Jane Street Capital — *September 2020 – January 2022*
Quantitative Trader — *London, UK*

Invited Talks

The Bunkbed Conjecture is Not Robust to Generalisation
2026 International Congress of Basic Science (ICBS), Beijing — *August 2026*

Hypercube Geodesics with Few Colour Changes
EPFL, Switzerland — *June 2026*

A Resolution of the Aharoni–Korman Conjecture
University of Cambridge — *October 2025*

Double-jump Phase Transition for the Reverse Littlewood–Offord Problem
London School of Economics — *February 2025*

Double-jump Phase Transition for the Reverse Littlewood–Offord Problem
University College London — *January 2025*

Tight Anti-Concentration of Rademacher Sums
University of Cambridge — *May 2024*

Teaching

Teaching assistant	Convex Geometric Analysis, Spring semester	*EPFL, 2026*
Supervisions	Part II Logic and Set Theory, Lent term	*University of Cambridge, 2025*
Supervisions	Part IA Numbers and Sets, Michaelmas term	*University of Cambridge, 2024*
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Supervisions	Part IA Numbers and Sets, Michaelmas term	*University of Cambridge, 2023*
Supervisions	Part IB Analysis and Topology, Michaelmas term	*University of Cambridge, 2022*

Technical skills

Programming Languages/Tools	Lean, Rust, OCaml, Java, Python, C#, LaTeX
Languages	English (native), French (intermediate)

Publications and preprints

The structure of FAC posets and the Aharoni–Korman conjecture *2026*
arXiv preprint

Hypercube geodesics with few colour changes *2026*
arXiv preprint

Uniformly balanced H-factors in multicoloured complete graphs *2026*
arXiv preprint
with A. Banerjee

Counterexamples to conjectures on strong maximality and minimality *2026*
Electronic Journal of Combinatorics
with B. Randall Shaw

Reverse Littlewood–Offord problems with parity conditions *2025*
arXiv preprint
with G. B. Sorkin

Connecting hypercube 1-factors *2025*
arXiv preprint
with B. Randall Shaw

Approximate Itai-Zehavi conjecture for random graphs *2025*
arXiv preprint (Random Structures & Algorithms, accepted)
with L. Lichev, A. Mond, J. Portier, and Y. Wang

Finding long cycles in percolated expander graphs *2026*
Random Structures & Algorithms

Monotonicity and decompositions of random regular graphs *2025*
arXiv preprint (Annals of Applied Probability, accepted)
with L. Lichev, A. Mond, J. Portier, and Y. Wang

Double-jump phase transition for the reverse Littlewood–Offord problem *2026*
Journal of the London Mathematical Society
with J. Portier and V. Souza

The Aharoni–Korman conjecture is false *2025*
arXiv preprint (Israel Journal of Mathematics, accepted)

A note on high-dimensional discrepancy of subtrees *2024*
arXiv preprint (Discrete & Computational Geometry, accepted)
with L. Lichev, A. Mond, and J. Portier

Discrepancies of spanning trees in dense graphs *2024*
arXiv preprint
with L. Lichev, A. Mond, and J. Portier

A uniform bound on almost colour-balanced perfect matchings in colour-balanced complete graphs *2025*
Electronic Journal of Combinatorics

Almost colour-balanced spanning forests in complete graphs *2024*
arXiv preprint
with A. Mond and J. Portier

The bunkbed conjecture is not robust to generalisation *2025*
European Journal of Combinatorics

On graphs with maximum difference between game chromatic number and chromatic number *2025*
Discrete Mathematics

On monotonicity in Maker-Breaker graph colouring games *2024*
Discrete Applied Mathematics

Tight Anti-Concentration of Rademacher Sums *2025*
Random Structures & Algorithms
with J. Portier

A note on interval colourings of graphs *2024*
European Journal of Combinatorics
with M. Axenovich, A. Girão, J. Portier, E. Powierski, M. Savery, Y. Tamitegama, and L. Versteegen

A new proof of the bunkbed conjecture in the $p \uparrow 1$ limit *2024*
Discrete Mathematics